



① Master formulae

$$\Gamma = \frac{\hbar}{\tau}$$

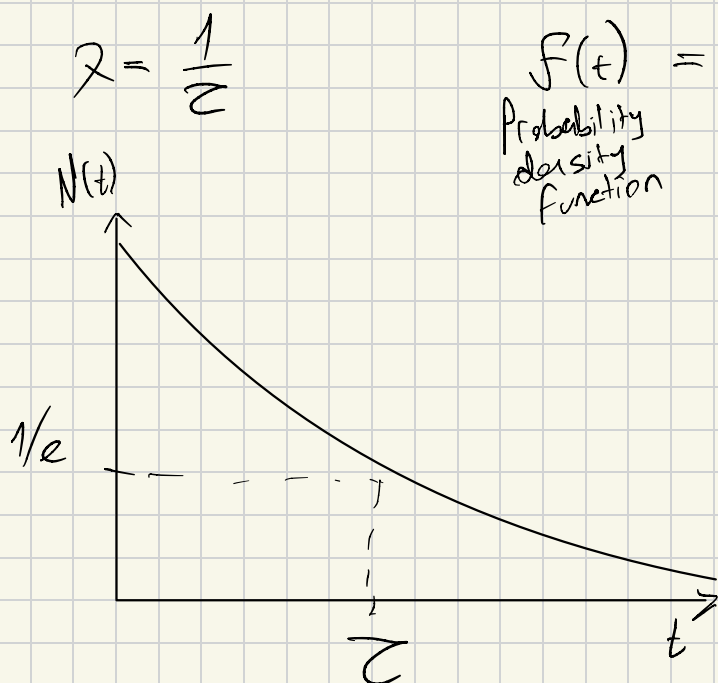
$N(t)$ a population of particles

$$\frac{dN(t)}{dt} = -\lambda N(t)$$

$$N(t) = N(0) e^{-\lambda t}$$

$$F(t) = \frac{1}{\tau} e^{-\frac{t}{\tau}}$$

Probability density function



What is Γ ?

natural width

It is the amplitude probability of the phenomenon. Let's take X decays

$$d\Gamma_X = \frac{1}{2M_X} \cdot \frac{1}{2S_X + 1} \sum_{\text{Sum over the final state spins}} M M^* d\Omega$$

Labels under the equation:
 - Under $\frac{1}{2M_X}$: average over spin
 - Under $\frac{1}{2S_X + 1}$: physics
 - Under $d\Omega$: phase space

$$\Gamma_{t \rightarrow \mu D \mu b} = \frac{G_F^2 \cdot M_E^5}{192 \cdot \pi^3}$$

$$d\Omega = \prod_{i \text{ over final state particles}} \frac{d^3 p_i}{(2\pi)^3 (2E_i)} (2\pi)^4 \delta\left[P_X - \sum_i p_i\right]$$

